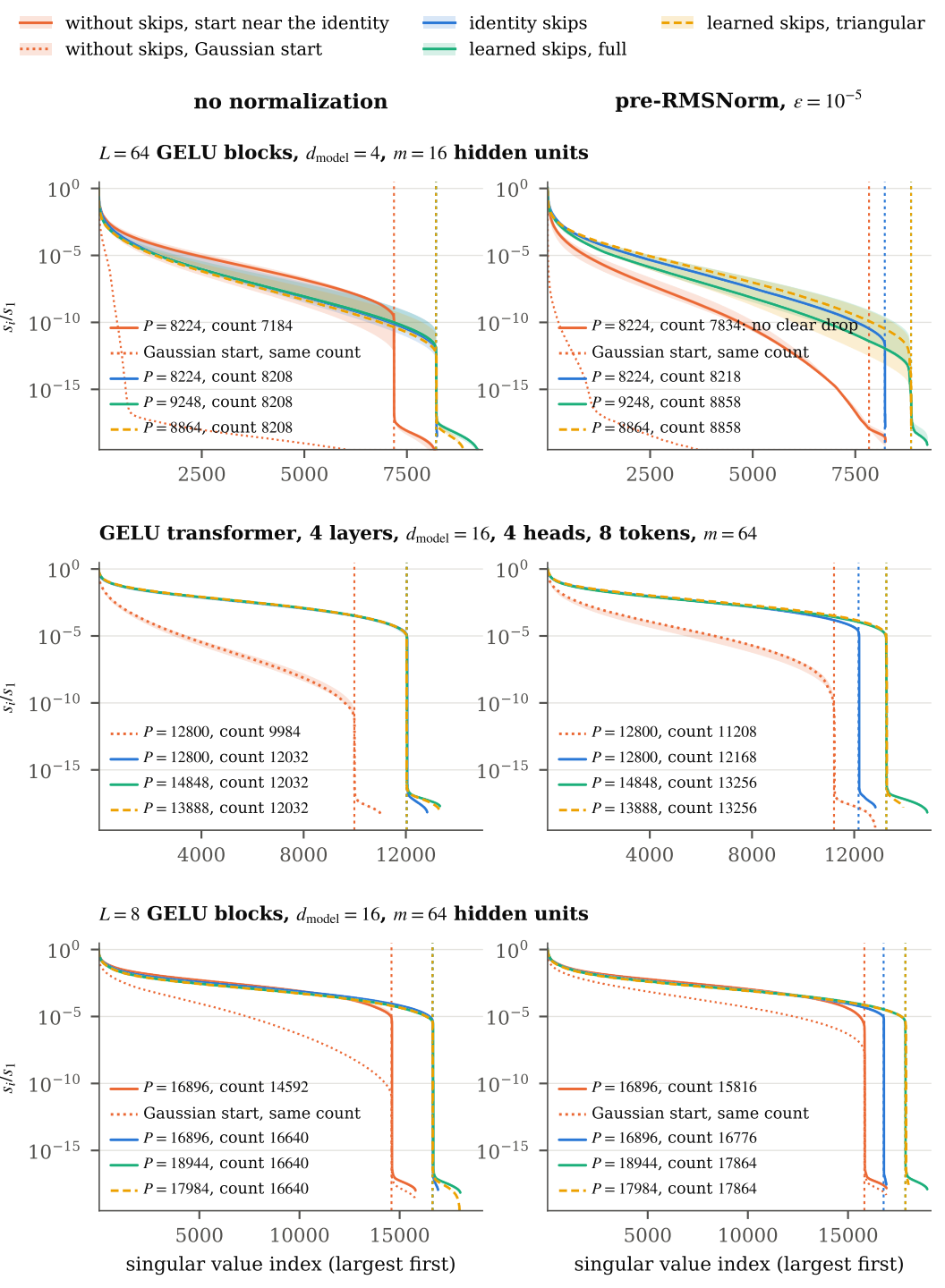

without skips, start near the identity
without skips, Gaussian start
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 64$ GELU blocks, $d_{\text{model}} = 4$, $m = 16$ hidden units
s_i/s_1
$P = 8224$, count 7184
Gaussian start, same count
$P = 8224$, count 8208
$P = 9248$, count 8208
$P = 8864$, count 8208
$P = 8224$, count 7834: no clear drop
Gaussian start, same count
$P = 8224$, count 8218
$P = 9248$, count 8858
$P = 8864$, count 8858
GELU transformer, 4 layers, $d_{\text{model}} = 16$, 4 heads, 8 tokens, $m = 64$
$P = 12800$, count 9984
$P = 12800$, count 12032
$P = 14848$, count 12032
$P = 13888$, count 12032
$P = 12800$, count 11208
$P = 12800$, count 12168
$P = 14848$, count 13256
$P = 13888$, count 13256
$L = 8$ GELU blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
$P = 16896$, count 14592
Gaussian start, same count
$P = 16896$, count 16640
$P = 18944$, count 16640
$P = 17984$, count 16640
$P = 16896$, count 15816
Gaussian start, same count
$P = 16896$, count 16776
$P = 18944$, count 17864
$P = 17984$, count 17864
singular value index (largest first)
singular value index (largest first)